(3)(b)

(1) MLE Parameter Estimation

assuming a gaussian distribution, likelihood function is,

$$p(X/w) = \prod_{n=1}^{m=N} \frac{1}{\sqrt{2\pi \sigma_n^2}} \exp\left(\frac{-\theta_n (t_n - W^T(x_n))^2}{2\sigma_n^2}\right)$$

to estimate parameters we find,

$$L(w / \phi(x_n)) = \underset{w}{\operatorname{argmax}} \prod_{n=1}^{n=N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(\frac{-\theta_n (t_n - W^T(x_n))^2}{2\sigma_n^2}\right)$$

taking log on both sides

$$\underset{w}{\operatorname{argmax}} \log L(w / \phi(x_n)) = \underset{w}{\operatorname{argmax}} \sum_{n=1}^{n=N} \left\{ -\frac{1}{2} \operatorname{Log}(2\pi\sigma_n^2) + \frac{1}{2}\left(-\frac{\theta_n (t_n - W^T(\phi(x_n)))^2}{\sigma_n^2}\right)\right\}$$

$$= \frac{1}{2} \underset{w}{\operatorname{argmax}} \sum_{n=1}^{N} \left\{ -\frac{\theta_n (t_n - W^T \phi(x_n))^2}{\sigma_n^2} - \operatorname{Log} 2\pi\sigma_n^2 \right\}$$

$$= \frac{1}{2} \underset{w}{\operatorname{argmin}} \sum_{n=1}^{N} \left\{ \frac{\theta_n (t_n - W^T \phi(x_n))^2}{\sigma_n^2} + \operatorname{Log}(2\pi\sigma_n^2) \right\}$$

this is the expression for MLE estimation

(ii) MAP estimation

$$P(W/X) = P(X/W) \cdot P(W)$$

taking log on both sides

$$\underset{W}{\operatorname{argmax}}\ P(W/X) = \underset{W}{\operatorname{argmax}} \log(P(X/W)) + \underset{W}{\operatorname{argmax}} \log(P(W))$$

I II

Now from eqn-① we know that

$$\underset{W}{\operatorname{argmax}}\ P(X/W) = \underset{W}{\operatorname{argmin}}\ \frac{1}{2}\left\{\frac{\sum_n (t_n - W^T(\phi(x_n)))^2}{\sigma_n^2} + \operatorname{Log}(2\pi\sigma_n^2)\right\}$$

→ also for II we assume a gaussian prior for the weight vector W as in 2(a)(ii)

$$P(W) = \frac{1}{\sqrt{(2\pi)^N |\Sigma_W|}} \exp\left(-\frac{1}{2}(W-\mu_W)^T \Sigma_W^{-1} (W-\mu_W)\right)$$

$$\Rightarrow \underset{W}{\operatorname{argmax}}\ (P(W/x)) = \underset{W}{\operatorname{argmin}}\ \frac{1}{2}\left\{\frac{\sum_n (t_n - W^T(\phi(x_n)))^2}{\sigma_n^2} + \operatorname{Log}(2\pi\sigma_n^2)\right\}$$

$$+ \underset{W}{\operatorname{argmax}}\left\{-\frac{1}{2}\operatorname{Log}(2\pi|\Sigma_W|) - \frac{1}{2}(W-\mu_W)^T \Sigma_W^{-1} (W-\mu_W)\right\}$$

$$= \underset{w}{\operatorname{argmin}} \; \frac{1}{2} \left\{ \frac{\theta_n \left(t_n - w^T(\phi(x_n))\right)^2}{\sigma_n^2} + \log\left(2\pi \sigma_n^2\right) \right\}$$

$$+ \underset{w}{\operatorname{argmin}} \; \frac{1}{2} \left\{ \log\left(2\pi |\Sigma_w|\right) + (w - \mu_w)^T \Sigma_w^{-1} (w - \mu_w) \right\}$$

assuming mean vector $\mu_w = 1$, the above equation reduces to

$$= \underset{w}{\operatorname{argmin}} \; \frac{1}{2} \left\{ \left(\frac{\theta_n \left(t_n - w^T(\phi(x_n))\right)^2}{\sigma_n^2} \right) + \log\left(2\pi \sigma_n^2\right) + w^T \Sigma_w^{-1} w + \log\left(2\pi |\Sigma_w|\right) \right\}$$

(Ans.)